

Search for Static and Oscillating EDMs in Storage Rings

Paolo Lenisa

University of Ferrara and INFN, Italy

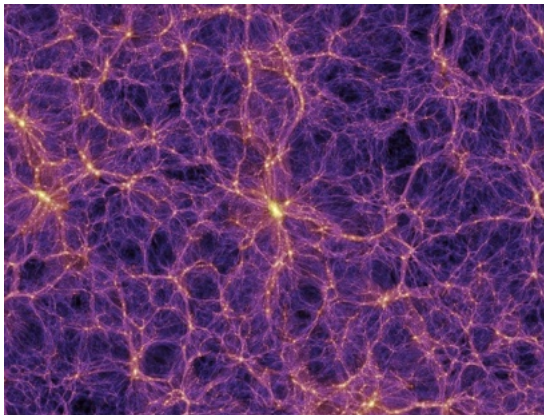
on behalf of the JEDI Collaboration

SPIN2025

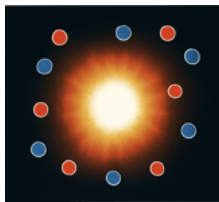
26th International Symposium on Spin Physics

Qingdao (China), September 26th, 2025

Motivation



Question 1: Why is our Universe Made of Matter?



Big-bang

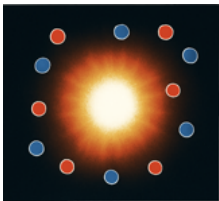
- Equal amounts of **matter** and **antimatter**



Now

- Universe dominated by matter: $\eta = \frac{n_B - n_{\bar{B}}}{n_\gamma}$
 - Measured: $\eta \approx 10^{-10}$
Bennet et al., *Astrophys. J. Suppl.* 148 (2003); Barger et al., *PLB* 566 (2003)
 - SM prediction: $\eta \approx 10^{-18}$
Bernreuther et al., *Lect. Notes Phys.* 591 (2002)

Question 1: Why is our Universe Made of Matter?



Big-bang

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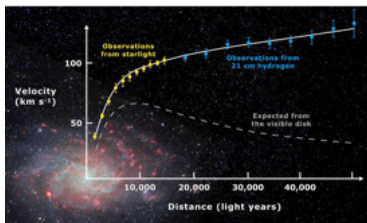
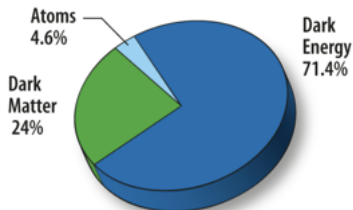
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Beyond the Standard Model

- Additional sources of CP violation beyond CKM mechanism required to explain the matter-antimatter asymmetry. [Sakharov, Soviet Physics Uspekhi 5 \(1991\)](#)

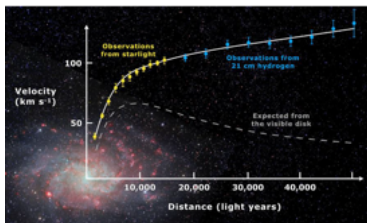
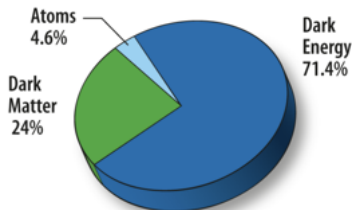
Question 2: What is the Nature of Dark Matter?



Experimental evidence

- Rotation curves of galaxies
Vera Rubin & Kent Ford, *The Astrophysical Journal*, 159 (1970)
- Gravitational lensing
Douglas Clowe et al, *Astroph. Journal Letters*, 648 (2006)
- Asymmetry in CMB
Planck Collaboration: *Astronomy & Astrophysics* 641, A6 (2020)

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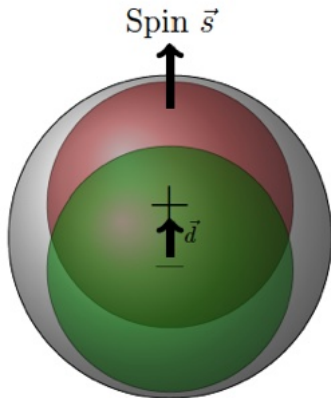
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What is Dark Matter made of?

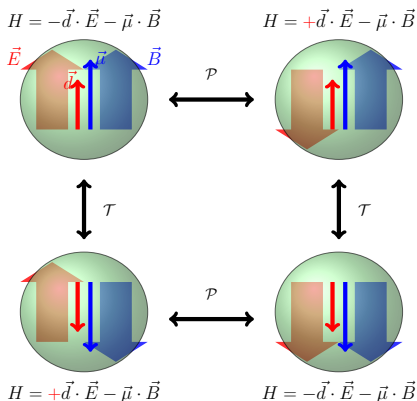
- WIMPs, Sterile neutrino, **Axions? Axion like particles?**
- Implies new physics beyond the Standard Model

Electric Dipole Moment (EDM)



- Permanent separation of positive and negative charge in the particle volume
- Fundamental particles property (like mag. moment, mass, charge)
- *Axion field* $\rightarrow$ *oscillating EDM*:
 $d = d_{DC} + d_{AC}\cos(\omega_a t + \phi_a); m_a c^2 = \hbar\omega_a$

$\mathcal{T}$ and $\mathcal{P}$ Violations in EDMs



$$H = -\mu \frac{\vec{s}}{s} \cdot \vec{B} - d \frac{\vec{s}}{s} \cdot \vec{E}$$

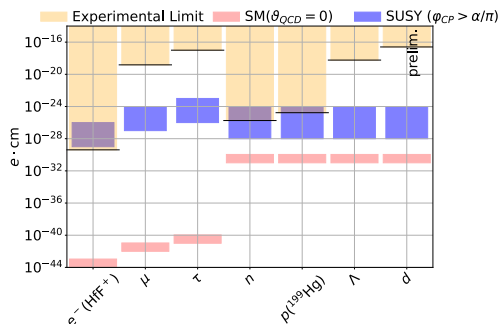
- $\mathcal{T}: H = -\mu \frac{\vec{s}}{s} \cdot \vec{B} + d \frac{\vec{s}}{s} \cdot \vec{E}$

- $\mathcal{P}: H = -\mu \frac{\vec{s}}{s} \cdot \vec{B} + d \frac{\vec{s}}{s} \cdot \vec{E}$

Implications of CP violation

- EDMs violate both $\mathcal{T}$ and $\mathcal{P}$ symmetries ($\stackrel{CPT}{=} \mathcal{CP}$)
- Nonzero EDM would indicate new CP-violating sources beyond SM
- Essential to explain matter-antimatter asymmetry

Static EDM Upper Limits



Current status

- No nonzero EDM has been measured so far

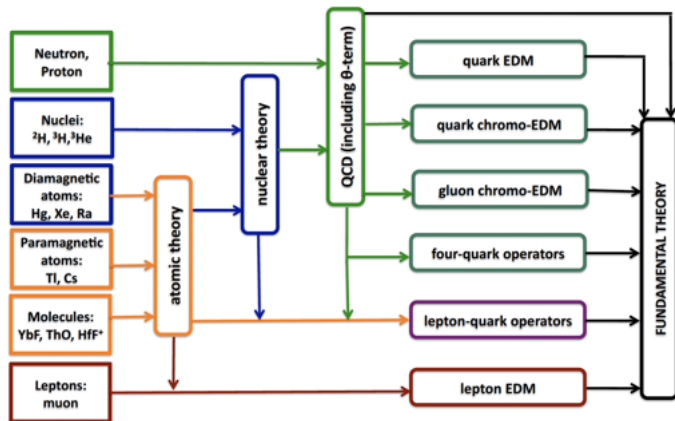
Direct measurements

- **Electron**: no direct measurement; strongest limit from HfF^+ molecule.
- **Proton**: no direct measurement; strongest limit from $^{199}_{80}\text{Hg}$.

Theoretical considerations

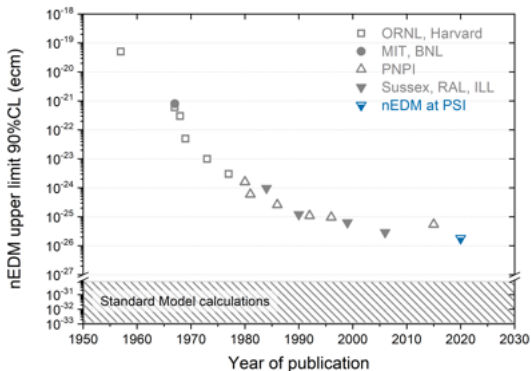
- EDM of single particle not sufficient to pinpoint the source of CP violation

Complementarity of EDM Measurements

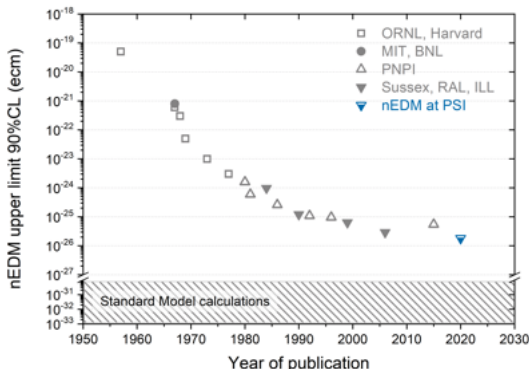


J. de Vries

Precision Takes Time: the Neutron EDM Limit



Precision Takes Time: the Neutron EDM Limit



Historical note

- Purcell and Ramsey (1950): *"The question of the possible existence of an electric dipole moment of a nucleus or of an elementary particle ... becomes a purely **experimental** matter"*
- First measurement of neutron EDM carried out in the early 1950s
- Published only in 1957, after Wu's discovery of PV in weak interaction

H.J. Smith, E.M. Purcell and N.F. Ramsey, Phys. Rev. 108, 1957

Axions and Axion-Like Particles (ALPs)

The Strong CP Problem

- QCD allows a CP-violating term: $\mathcal{L}_\theta = \theta \frac{g_s^2}{32\pi^2} G_{\mu\nu}^a \tilde{G}^{a\mu\nu}$
- This would induce a large neutron EDM, but experiments find it $\simeq 0$.
- $\Rightarrow$ Implies an extremely small θ : the **strong CP problem**.

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Peccei-Quinn Mechanism and Axions

- Peccei-Quinn dynamically drives $\theta \rightarrow 0$ [Peccei, Quinn, PRL 38 \(1977\)](#);
- Predicts a new pseudoscalar particle: the **axion** [Wilczek, PRL 40 \(1978\)](#)
- Generalizations $\Rightarrow$ **Axion-Like Particles (ALPs)** [Kim, PRL 43 \(1979\)](#); [Dine et al, PLB 104 \(1981\)](#)

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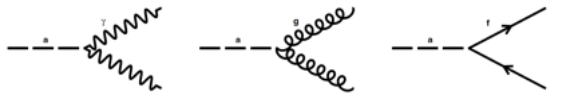
Connection to Dark Matter

- Axion/ALPs are excellent dark-matter candidates:
 - ▶ Very light, very stable; extremely weakly interacting

Experimental Searches

- Large effort to detect axion/APLs
 - ▶ Haloscopes (galactic DM axions); [P. Sikivie \(1983\)](#); CAST-CAPP, CAST-RADES, ...
 - ▶ Helioscopes (solar axions); [P. Sikivie, CAST](#); IAXO ...
 - ▶ Light-shining-trough wall experiments (photon coupling); [P. Sikivie](#); ALPS; CROWS..

Axion Coupling - Axion Induced Oscillations

$$\mathcal{L} : -\frac{\alpha}{8\pi} \frac{C_\gamma}{f_a} \mathbf{a} F_{\mu\nu} \tilde{F}^{\mu\nu} \quad -\frac{\alpha_s}{8\pi} \frac{C_G}{f_a} \mathbf{a} G_{\mu\nu}^b \tilde{G}^{b,\mu\nu} \quad -\frac{1}{2} \frac{C_N}{f_a} \partial_\mu \mathbf{a} \bar{\psi}_f \gamma^\mu \gamma^5 \psi_f$$


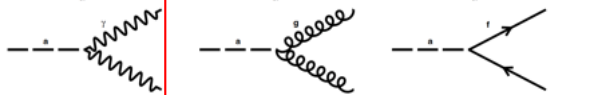
oscillating
Electric Dipole Moment (oEDM) axion wind term

Axion-field description

- For low axion masses, if axions saturate dark matter they can be described as:

$$\mathbf{a}(t) = a_0 \cos(\omega_a(t) + \phi_a); m_a c^2 = \hbar \omega_a; \text{coupling scales as } \propto \frac{1}{f_a} \propto m_a$$

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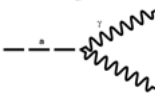
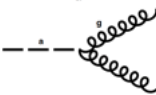
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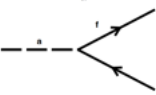
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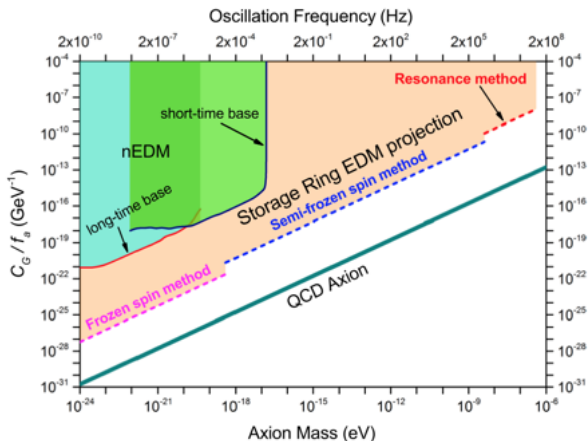
ALP-induced spin-interaction

- Gluon fields $\Rightarrow$ *oscillating EDMs*; [Graham et al., PRD 84 \(2011\)](#)
- Fermion coupling with axion field gradient (pseudomagnetic field)
 $\Rightarrow$ *axion wind effect*; [Graham et al., PRD 88 \(2013\)](#)

Measurement concept in a storage ring

- Use spin precession of polarized beam in a storage ring
- Look for resonance signatures

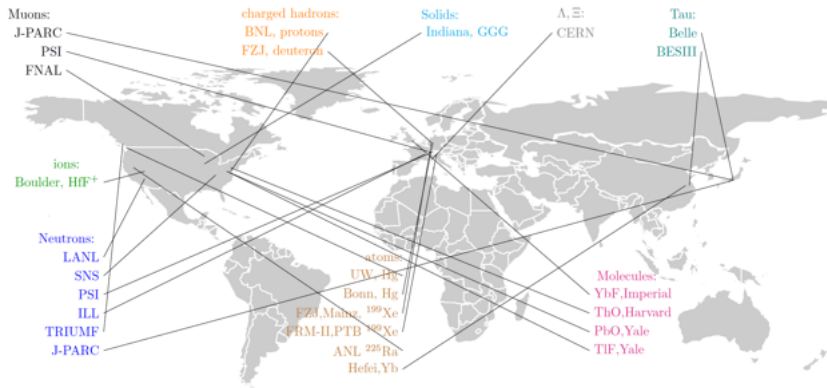
Axion Search with Storage Ring EDM Method



S. P. Chang et al. Phys. Rev. D 99, 083002

- Experimental limits on axion-gluon coupling from oscillating EDM searches.

EDM Experiments/Activities around the World

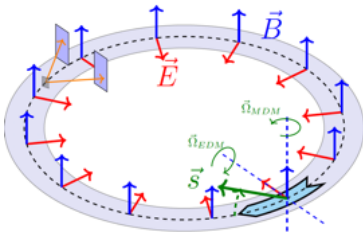
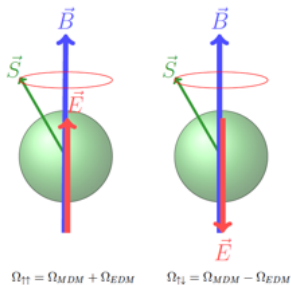


World Map of Current Experiments on Wavy Dark Matter

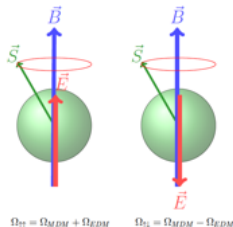


C. O'Hare, doi:10.5281/zenodo.3932430, <https://github.com/cajohare/AxionLimits>

Experimental methods



Measuring the EDM of Neutral Particles



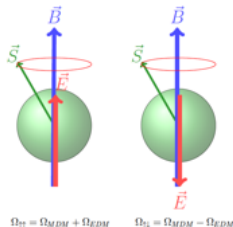
Spin Precession in **E** and **B** Fields

$$\vec{\Omega} = \Omega_{\text{MDM}} \pm \Omega_{\text{EDM}} = \frac{\mu \mathbf{B} \pm d \mathbf{E}}{|\vec{S}|}$$

Orders of Magnitude for $d_n = 1 \times 10^{-26} \text{ e} \cdot \text{cm}$

- In B_{earth} : $\Omega \approx 9,000 \text{ s}^{-1}$
- In $E = 10^7 \text{ V/m}$: $\Omega_{\text{EDM}} \approx 3 \times 10^{-6} \text{ s}^{-1}$

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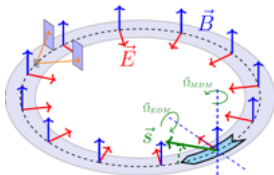
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Measuring the EDM of Charged Particles

More challenging $\Rightarrow$ requires dedicated storage rings.

Spin-Precession of Charged Particles in Storage Rings



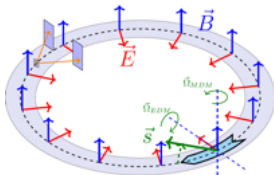
- Equation of motion for spin vector $\vec{S}$ in the rest frame of the particle:

$$\frac{d\vec{S}}{dt} = \vec{\Omega} \times \vec{S} = \underbrace{\vec{\mu}}_{\text{blue}} \times \vec{B} + \underbrace{\vec{d}}_{\text{red}} \times \vec{E}$$

- Spin-precession relative to the direction of motion:

$$[(\underbrace{\vec{\Omega}_{MDM}}_{\text{blue}} + \underbrace{\vec{\Omega}_{EDM}}_{\text{red}}) - \vec{\Omega}_{cycl}] = \frac{-q}{m} \left[\underbrace{G\vec{B} + \left(G - \frac{1}{\gamma^2 - 1}\right) \vec{v} \times \vec{E}}_{=\Omega_{MDM} - \Omega_{cycl}} + \underbrace{\frac{\eta}{2} (\vec{E} + \vec{v} \times \vec{B})}_{=\Omega_{EDM}} \right]$$

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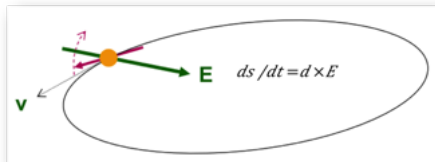
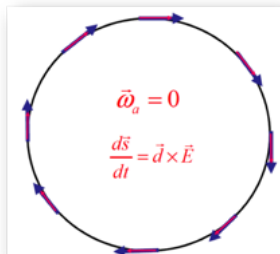
Frozen spin

- $\vec{\Omega}_{MDM} - \vec{\Omega}_{cycl} = 0 \Rightarrow$ frozen spin (momentum and spin stay aligned)
 - Achievable with **pure electric field** for proton ($G > 0$): $G = \frac{1}{\gamma^2 - 1}$

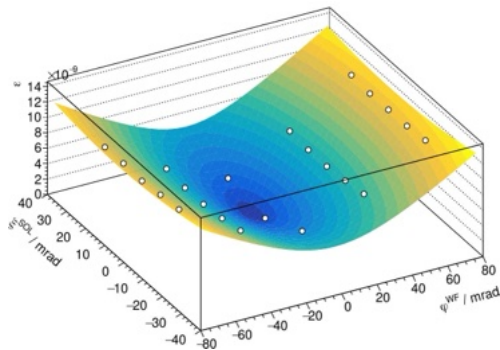
Search for Static EDM in Storage Rings

Storage ring method to measure EDM of charged particle

- 1 Inject beam of polarized particles in storage ring
- 2 Align spin along momentum ($\rightarrow$ freeze horiz. spin-precession)
- 3 Search for time development of vertical polarization



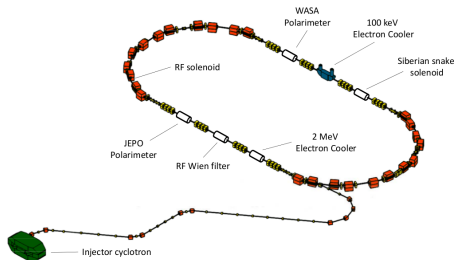
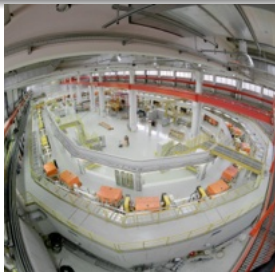
Achievements at COSY Storage Ring



The COSY Storage Ring at FZ-Jülich (Germany)

COoler SYnchrotron COSY

- Cooler and storage ring for (pol.) protons and deuterons.
- Momenta $p = 0.3\text{--}3.7$ GeV/c
- Phase-space cooled internal and extracted beams

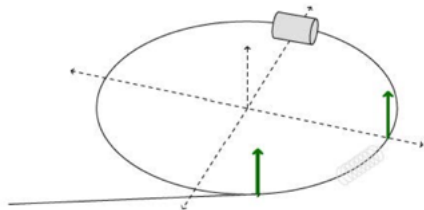


Previously used as spin-physics machine for hadron physics:

- Ideal starting point for Storage Ring EDM related R&D
- Dedicated and unique experimental effort worldwide
- Closed end 2023: essential R&D expts. with MAGNETIC ring successfully done.

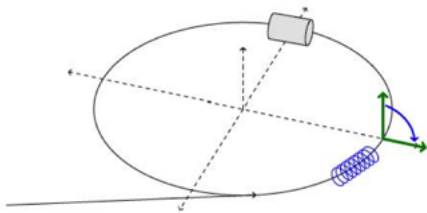
Experiment Preparation

- 1 Inject and accelerate vertically pol. deut. to $p \approx 1$ GeV/c



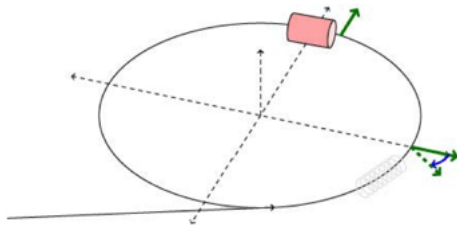
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- 2 Flip spin with solenoid into horizontal plane



Experiment Preparation

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- 2 Flip spin with solenoid into horizontal plane
- 3 Extract beam slowly (100 s) on Carbon target
- 4 Measure asymmetry and determine spin precession



Prerequisite: Polarimetry

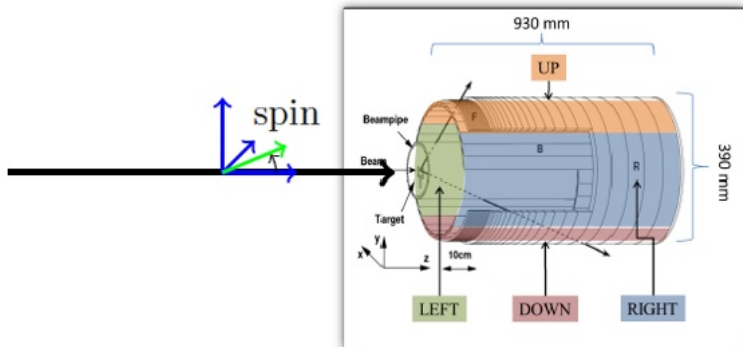
Spin-dependent elastic deuteron-carbon scattering

- Up/Down asymmetry $\propto$ horizontal polarization

▶ $N_{up,down} \propto 1 \pm \frac{3}{2} p_z A_y \sin(\nu_s \omega_{rev} t);$

$\nu_s = \gamma G \simeq -0.161$ (spin-tune); $f_{rev} = 781$ kHz; $f_s = 126$ kHz.

- Left/Right asymmetry $\propto$ vertical polarization $\rightarrow$ d



Time-Stamp System

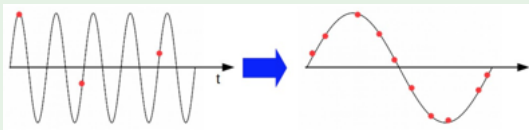
Asymmetry: $\epsilon = \frac{N_{up} - N_{down}}{N_{up} + N_{down}} = p_z A_y \sin(2\pi \cdot \nu_s \cdot n_{turns})$

Challenge

- Spin precession frequency: 126 kHz
- event rate: $5000 \text{ s}^{-1} \rightarrow 1 \text{ hit} / 25 \text{ precessions} \rightarrow \text{no direct fit of rates}$

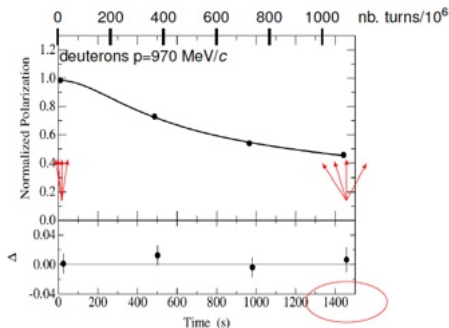
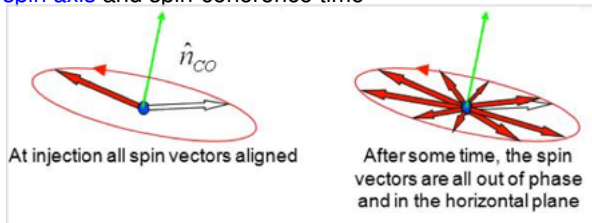
Solution: map many event to one cycle

- Counting turn number $n \rightarrow$ phase advance $\phi_s = 2\pi\nu_s n$
- For intervals of $\Delta n = 10^6$ turns: $\phi_s \rightarrow \phi_s \bmod 2\pi$



Prerequisite: Long Spin-Coherence Time

- Invariant spin axis and spin-coherence time

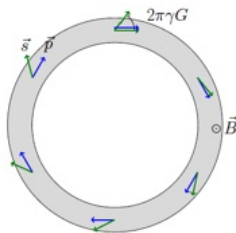


1st Major Achievement

[Phys. Rev. Lett. 117 (2016) 054801]

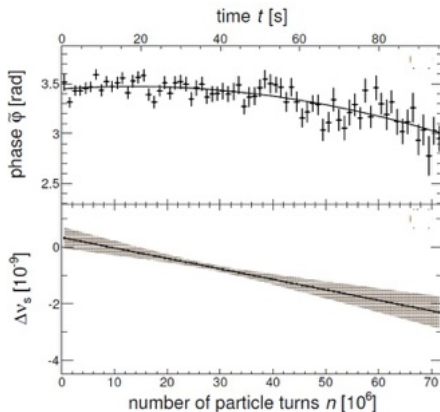
- $\tau_{SCT} = (1173 \pm 172)$ s
- Previously: $\tau_{SCT}(\text{VEPP}) \approx 0.5$ s
- SCT is of crucial importance, since
$$\sigma_{STAT} \propto \frac{1}{\tau_{SCT}}$$

Precise Determination of the Spin-Tune



Spin-tune ν_s

$$\nu_s = \gamma G = \frac{\text{nb. spin-rotations}}{\text{nb. particle-revolutions}}$$



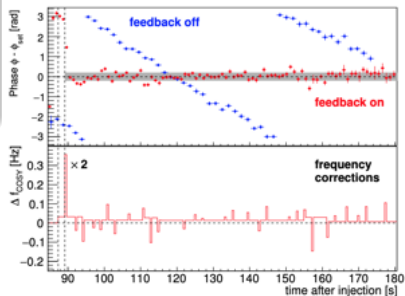
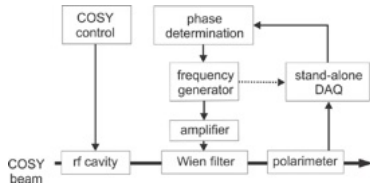
2nd major achievement [*Phys. Rev. Lett.* 115 (2015) 094801]

- Interpolated spin tune in 100 s:
- $|\nu_s| = (16097540628.3 \pm 9.7) \times 10^{-11}$ ($\Delta\nu_s/\nu_s \approx 10^{-10}$)
- → new tool to study systematic effects in storage rings

Phase-Locked Spin Precession

Spin-feedback system maintains:

- resonance frequency
- phase between spin-precession and device RF



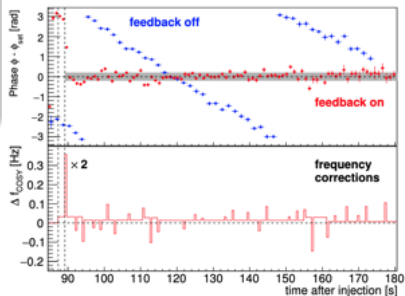
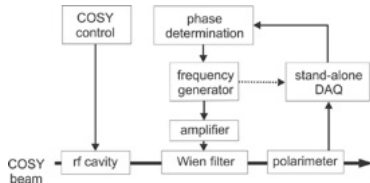
3rd major achievement [*Phys. Rev. Lett.* 119 (2017) 014801]:

Error of phase-lock $\sigma_{\phi} = 0.21$ rad

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- phase between spin-precession and device RF



3rd major achievement [*Phys. Rev. Lett.* 119 (2017) 014801]:

Error of phase-lock $\sigma_\phi = 0.21$ rad

At COSY freezing of spin precession not possible
→ **phase-locking** required to achieve precision for EDM

Measurement of EDM in a magnetic ring

First-ever direct EDM measurement using this method

Concept

- Radial field $\vec{E} = c \vec{\beta} \times \vec{B}$ from relativistic motion in vertical $\vec{B}$
- Spin precession: $\frac{d\vec{S}}{dt} \propto \vec{d} \times \vec{E}$

Measurement of EDM in a magnetic ring

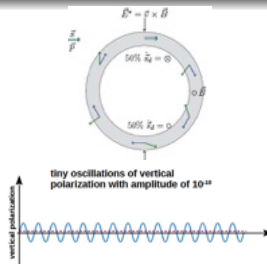
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Problem: $\Rightarrow$ only small oscillation of vertical polarization p_y due to EDM.

- Momentum $\uparrow$, spin $\uparrow \Rightarrow$ spin kicked up
- Momentum $\uparrow$, spin $\downarrow \Rightarrow$ spin kicked down
- $\Rightarrow$ no accumulation of vert. asymmetry



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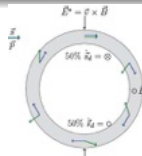
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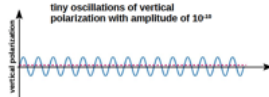
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- Momentum $\uparrow$, spin $\downarrow \Rightarrow$ spin kicked down
- $\Rightarrow$ no accumulation of vert. asymmetry

Solution: RF-Wien filter

- Lorentz force: $\vec{F}_L = q(\vec{E} + \vec{v} \times \vec{B}) = 0$
 $\rightarrow$ particle trajectory not affected
- $\vec{B} = (0, B_y, 0)$ and $\vec{E} = (E_x, 0, 0)$
 $\rightarrow$ mag. moment influenced

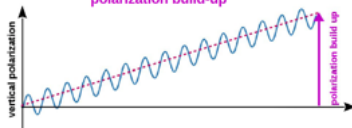


tiny oscillations of vertical polarization with amplitude of 10^{-13}

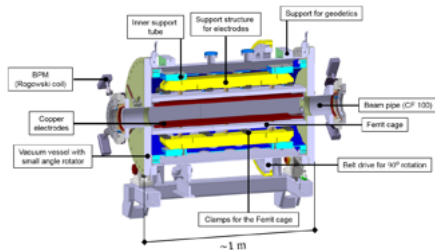


phase lock between spin precession and RF Wien filter

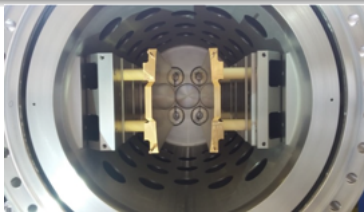
polarization build-up



Measurement of EDM in a magnetic ring RF-Wien filter¹



- Waveguide provides $\vec{E} \times \vec{B}$ by design.
- Minimal $\vec{F}_L$ by careful electromagnetic design of all components.



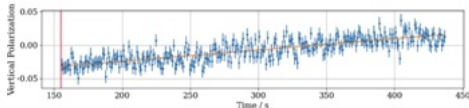
¹Joint development with RWTH Aachen

Pilot Bunch Comagnetometer

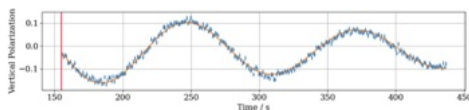
4th major achievement [*Phys. Rev. Research* 7, 023257]

- Observation of $p_y(t)$ with two stored bunches: pilot bunch and signal bunch
 - ▶ Pilot bunch shielded from Wien-filter RF by fast RF switches

- Pilot bunch

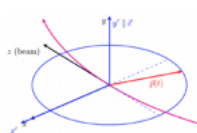


- Signal bunch

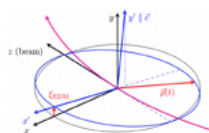


- No oscillations in pilot bunch.
- Decoherence visible in signal bunch.

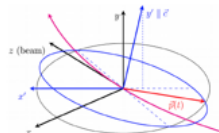
Tilting of the Invariant Spin Axis



EDM absence



EDM effect



Magnetic misalignm.

$$\phi_{EDM} = \arctan\left(\frac{\eta_{EDM}\beta}{2G}\right)$$

$$\vec{n}_{ISA} \approx \begin{pmatrix} \phi_{EDM} + \phi_{Ring} \\ 1 \\ \xi_{Ring} \end{pmatrix}$$

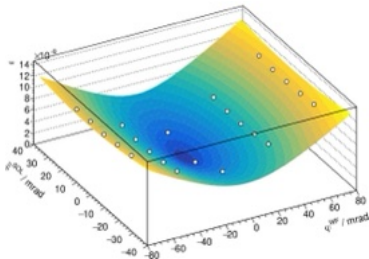
EDM + magnetic misalignments tilt the invariant spin axis

- Presence of EDM $\rightarrow \phi_{EDM} > 0$
- Presence of magnetic misalignments $\rightarrow \phi_{EDM} \text{ \& } \xi_{ring} > 0$
 - ▶ $\rightarrow$ spin precess around the $\vec{n}_{ISA}$ axis
 - ▶ $\rightarrow$ oscill. vert. polarization $p_y(t)$

Results of dEDM Precursor Experiment

EDM resonance strength map for ϵ^{EDM}

- Includes tilts of invariant spin axis due to EDM and magnetic ring imperfections.



Preliminary result on static EDM - 1st ever measurement of deuteron's EDM

- Determination of minimum via fit with theoretical surface function yields:

- $n_x^{WF} \text{ (mrad)} = -2.1(12)$

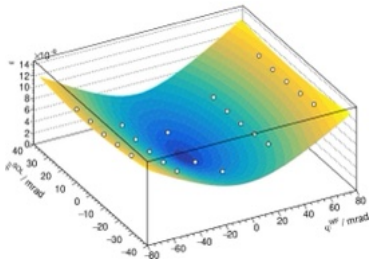
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$$\Rightarrow d_{EDM} < 3 \cdot 10^{-17} e \cdot cm \text{ (95 \% CL)}$$

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$$\Rightarrow d_{EDM} < 3 \cdot 10^{-17} e \cdot cm \text{ (95 \% CL)}$$

- Only other direct measurement: $\mu_{EDM} < 1.9 \cdot 10^{-19} e \cdot cm \text{ (95 \% CL)}$

→ see talk of A. Andres on 23.09 for details

Search for Axion-Like Particles in a Storage Ring

First-ever search using this method

Axions and oscillating EDM

- Axion interaction with ordinary matter: $\frac{a}{f_0} F_{\mu\nu} \tilde{F}_{\mu\nu}$, $\frac{a}{f_0} G_{\mu\nu} \tilde{G}_{\mu\nu}$, $\frac{\partial_\mu a}{f_a} \bar{\Psi} \gamma^\mu \gamma_5 \Psi$
- $\frac{a}{f_0} G_{\mu\nu} \tilde{G}_{\mu\nu} \rightarrow$ coupling to gluons with same structure as QCD- θ term
- Generation of an oscillating EDM with freq. related to mass: $\hbar\omega_a = m_a c^2$

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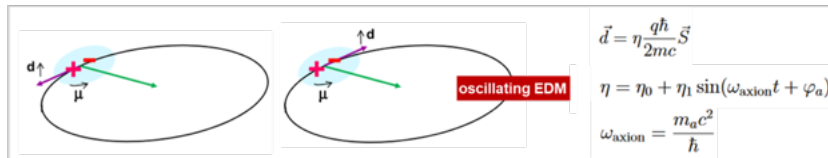
First-ever search using this method

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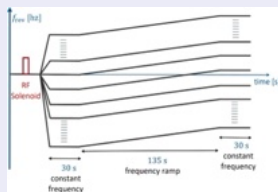
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Experimental approach

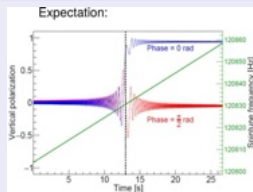
- Mag. dipole moment (MDM) $\rightarrow$ spin prec. in B field $\rightarrow$ nullifies static EDM effect
- Osc. EDM resonant condition ($\omega_a = \omega_s$) $\rightarrow$ buildup of out-of-plane spin rotation



Momentum ramps (f_{rev}) searching for polarization changes

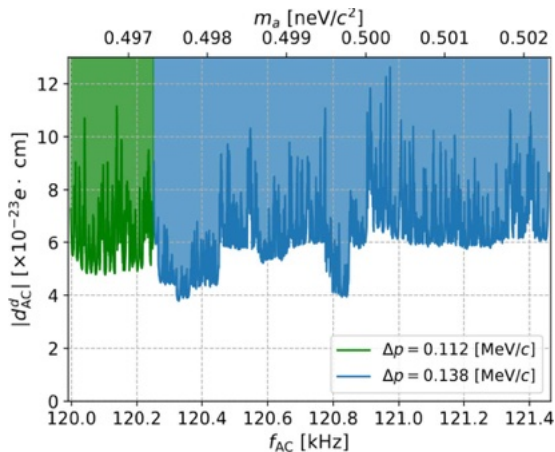


- Organization of frequency ramps.



- Jump of vertical polarization when resonance is crossed, for $\omega_a = \omega_s$

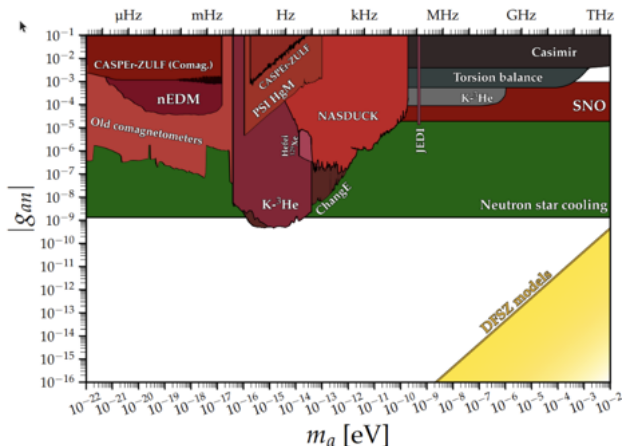
Bound on the Oscillating EDM of the Deuteron



Observed oscillation amplitudes from 4 bunches

- 90 % CL upper limit on the ALPs induced oscillating EDM
- Average of individual measured points $d_{AC} < 6.4 \times 10^{-23} e \cdot cm$

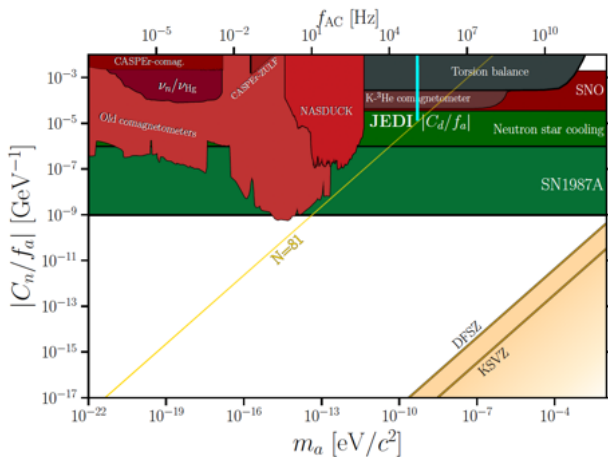
Axion Coupling to EDM Operator $g_{ad\gamma}$ (e.g. Axion/Gluon Coupling)



Limits on axion/ALP neutron coupling from the Particle Data Group

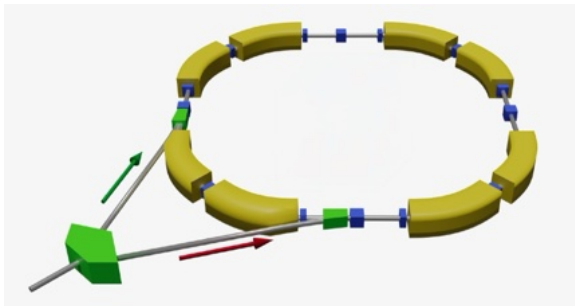
- $g_{ad\gamma} = d_{AC} = \frac{d_{AC}}{a_0}$; $a_0 = 0.55 \frac{\text{GeV}}{\text{cm}^3}$ (Dark Matter saturated by ALPs)
- It includes JEDI result: [S. Karanth et al., Phys. Rev. X 13 \(2023\) 031004](#)

Axion Wind Effect: Coupling to Nucleons $\frac{C_N}{f_a}$



- Storage ring experiments particularly sensitive to axion wind effect

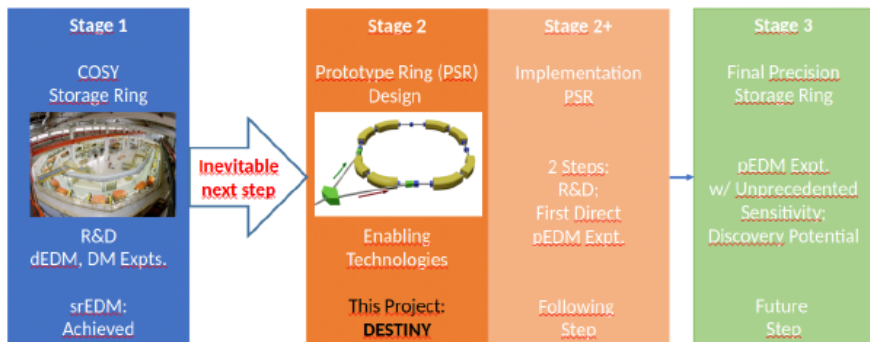
Next steps



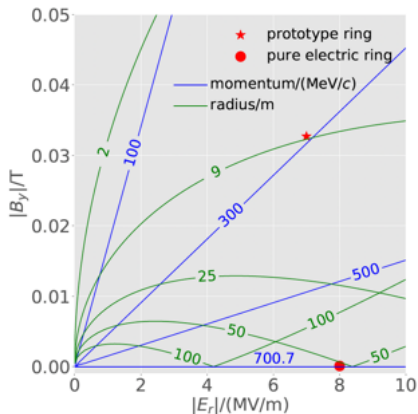
Objective: Construction of a Dedicated SR for EDM Studies

Possible approaches

- One step approach: immediate construction of final ring
- Staged approach: intermediate prototype ring



Design Options for Frozen-Spin Proton Rings



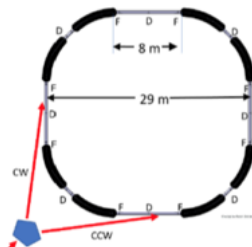
Two-options:

- **Hybrid - 100 m ring:**
p = 300 MeV/c
bending radius ≈ 9 m at $E=7$ MeV/m
- **Pure electric - 800 m ring;**
p = 707 MeV/c;
bending radius ≈ 50 m at $E = 8$ MeV/m

Stage 2: Prototype EDM Storage Ring

100 m circumference

- p at 30 MeV **all-electric** CW-CCW beams operation
- Frozen spin including additional **vertical magnetic fields**



Challenges

- All electric & E-B combined deflection
- Storage time
- **CW-CCW operation** → next slide
 - ▶ Orbit control
 - ▶ Control of orbit difference
- Polarimetry
- Spin-coherence time
- Magnetic moment effects
- Stochastic cooling

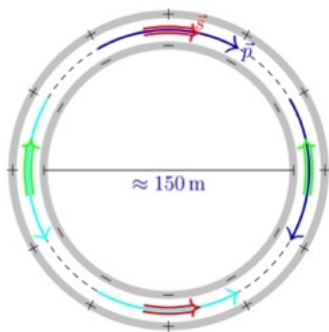
Objectives of PSR

- Study open issues.
- First direct proton EDM measurement.

Stage 3: Precision EDM Ring

500 m circumference (with $E = 8 \text{ MV/m}$)

- All-electric deflection
- Magic momentum for protons ($p = 707 \text{ MeV/c}$)



Challenges

- All-electric deflection
- Simultaneous CW/CCW beams
- Phase-space cooled beams
- Long spin coherence time ($> 1000 \text{ s}$)
- Non-destructive precision polarimetry
- Optimum orbit control
- Optimum shielding of external fields
- Control of residual B_r fields

"Holy Grail" storage ring (largest electrostatic ever conceived)

Developm.: Extending the Spin-Coherence Time in Storage Rings

Spin-Coherence Time (SCT)

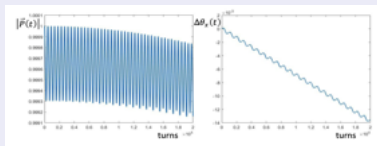
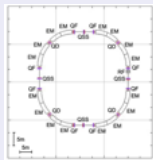
- Polarization vector: $\vec{P}(t) = \frac{1}{n} \sum_{i=1}^n \vec{s}_i(t)$
- Definition: $P(\tau) = P_0/e$

Analytical Model

- $\tau \propto 1/\Delta\nu_s$ (spin-tune spread)
- $\Delta\nu_s \propto \Delta L/L$ (path lengthening)
- $\Rightarrow$ Optimization via sextupole families

$$\begin{aligned}\frac{\Delta L}{L} &= -\frac{\pi}{L}(\epsilon_x \xi_x + \epsilon_y \xi_y) + \alpha_0 \delta + \alpha_1 \delta^2 \\ &= \alpha_1 \delta^2 - \frac{\pi}{L} \epsilon_x \xi_x - \frac{\pi}{L} \epsilon_y \xi_y\end{aligned}$$

Validation through Bmad Simulations



$\Rightarrow$ see talk of R. Shankar on 23.09 for details

Statistical Reach of the EDM Ring

High precision, primarily electric storage ring

- Beam intensity: $N=4 \cdot 10^{10}$ per fill
- Polarization: $P=0.8$
- Spin coherence time: $\tau = 1000$ s
- Electric fields: $E = 8$ MV/m
- Polarimeter analyzing power: $A = 0.6$
- Polarimeter efficiency: $f = 0.005$

Expected statistical sensitivity in 1 year of DT:

- $\sigma_{stat} = \frac{2\hbar}{\sqrt{N} f \tau P A E} \Rightarrow \sigma_{stat} = 2.4 \cdot 10^{-29} e \cdot cm$
- $\Rightarrow$ Challenge: get σ_{syst} to the same level.

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B. Marciano (Snowmass Workshop, Sept. 2020) about $d_p \sim 10^{-29}$:

- $d_p \sim \frac{e \cdot m}{\Lambda_{NP}^2} \sin \phi^{NP}$ ($\Lambda_{NP} \equiv$ scale of NP; $\phi^{NP} \equiv$ complex CP violation phase of NP)
- If ϕ^{NP} is of $O(1) \Rightarrow \Lambda_{NP} \sim 3000$ TeV probed
- If $\Lambda_{NP} \sim O(1$ TeV) $\Rightarrow \phi^{NP} \sim 10^{-6}$ probed

Systematic Limits and Mitigation

- Signal: $\Omega_{EDM} = \frac{dE}{s\hbar} = 2.4 \cdot 10^{-9} s^{-1}$ for $d = 10^{-29} e \cdot cm$

Possible systematic contributions

- Radial B-field: $B_r = 10^{-17} T$: $\Omega_{B_r} = 1.7 \cdot 10^{-9} s^{-1}$
- Geometric phase: $B_{long} = B_{vert} = 10^{-9} T$: $\Omega_{B_r} = \frac{eGB_r}{16m} \frac{1}{f_{rev}} = 3.7 \cdot 10^{-9} s^{-1}$
- General relativity: $\Omega_{GR} = - \frac{\gamma}{\gamma^2+1} \frac{\beta g}{c} = - 4.4 \cdot 10^{-8} s^{-1}$

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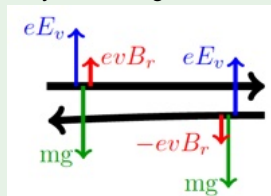
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Control strategy

- Use of two beams running clockwise (CW) and counterclockwise (CCW):
 $\Omega_{CW} = \Omega_{EDM} + \Omega_{GP} + \Omega_{GR} + \Omega_{B_r}$; $\Omega_{CCW} = \Omega_{EDM} - \Omega_{GP} - \Omega_{GR} + \Omega_{B_r}$
 $\Rightarrow$ In the sum, $\Omega_{GP} + \Omega_{GR}$ cancel out.
 $\Rightarrow$ Effect of B_r removed by observing relative displacement of the two beams.



J. Pretz

Conclusions

EDM searches in Storage Rings

- EDM searches probe new CP violation $\Rightarrow$ key to matter-antimatter asymmetry
- Key developments in accelerator technology

Fundamental achievements at COSY

- Spin-control tools
- First measurement of (static and oscillating) deuteron EDM

Next step

- Design and construction of a *pure electrostatic* EDM proton ring
- Possible approaches
 - ▶ Direct approach
 - ▶ Staged approach

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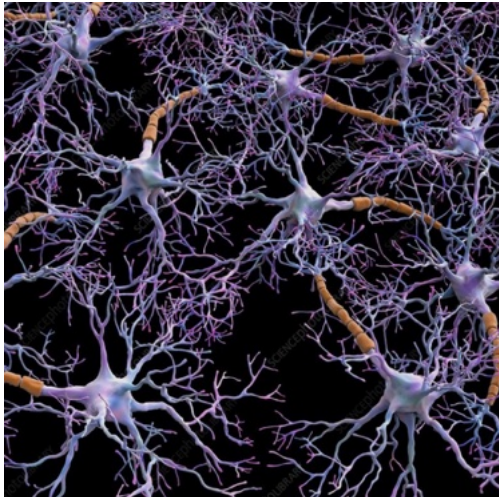
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Outstanding discovery potential!



Thank you!

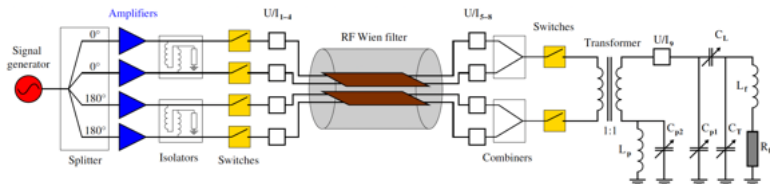
JEDI Collaboration: selected publications

- D. Eversmann et al (JEDI Collaboration): [New method for a continuous determination of the spin tune in storage rings and implications for precision experiments](#) - Phys. Rev. Lett. 115, 094801 (2015)
- J. Slim, et al.: [Electromagnetic simulation and design of a novel waveguide rf-Wien filter for electric dipole moment measurements of protons and deuterons](#) - Nucl. Instr. and Meth. A: 828, 116 (2016), ISSN 0168-9002
- G. Guidoboni et al. (JEDI Collaboration): [How to reach a thousand-second in-plane polarization lifetime with 0.97 GeV/c deuterons in a storage ring](#) - Phys. Rev. Lett. 117, 054801 (2016)
- N. Hempelmann et al. (JEDI Collaboration): [Phase locking the spin precession in a storage ring](#) - Phys. Rev. Lett. 119, 014801 (2017)
- F. Abusaif (CPEDM Collaboration): [Storage Ring to Search for Electric Dipole Moments of Charged Particles - Feasibility Study](#) - (CERN, Geneva, 2021)
- S. Karanth et al. (JEDI Collaboration): [First Search for Axion-Like Particles in a Storage Ring Using a Polarized Deuteron Beam](#) - S. Karanth et al., Phys. Rev. X 13 (2023) 031004.
- J. Slim, et al. (JEDI Collaboration): [Proof-of-principle demonstration of a pilot bunch comagnetometer in a stored beam](#) - J. Slim et al., Phys. Rev. Research 7, 023257

Spare slides

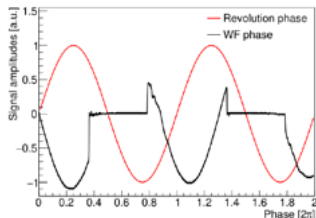
Implementation of fast switches² at RF Wien filter

Modification of driving circuit



GaN HEM FET-based solution:

- Short switch on/off times ($\approx$ few ns).
- High power capabilities ($\approx$ few kV).
- On board power damping (-30 dB)
- Symmetric switch on/off times ($\approx$ ns).



Switches

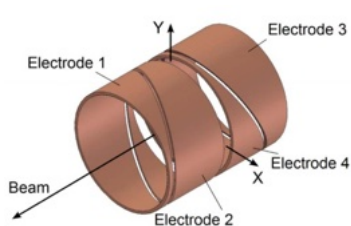
- Capable to handle up to 200 W each
- Permits system to run near a total power of 0.8 kW in pulsed mode

¹ Developed together with Fa. barthel HF-Technik GmbH, Aachen

Measurement of EDM in a magnetic ring

Beam position monitors for srEDM experiments

- Main adv.: short install. length (≈ 1 cm in beam direction)



Conventional BPM

- Easy to manufacture
- Length = 20 cm
- Resolution $\approx 10 \mu\text{m}$

Rogowski BPM (warm)

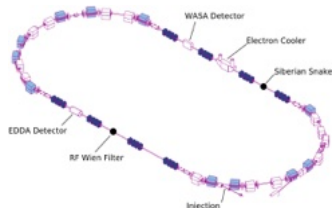
- Excellent RF-signal response
- Length = 1 cm
- Resolution $\approx 1.25 \mu\text{m}$

- 2 coils installed at entrance and exit of RF Wien filter

Strength of EDM resonance

EDM induced polarization oscillation

- Described by: $p_y(t) = a \sin(\Omega^{py} t + \phi_{RF})$
- EDM resonance strength: ratio of Ω^{py} to orbital ang. frequency Ω^{rev} : $\epsilon^{EDM} = \frac{\Omega^{py}}{\Omega^{rev}}$



Methodology of EDM measurement

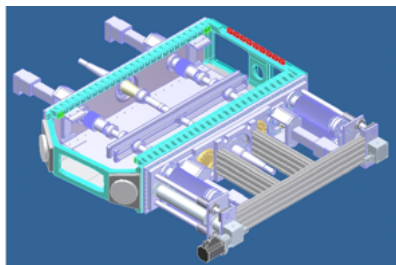
Two features simultaneously applied in the ring:

- 1 RF Wien-filter rotated by a small angle $\rightarrow$ generates small radial magnetic RF-field $\rightarrow$ affects the spin evolution.
- 2 In addition: longitudinal magnetic field in ring opposite to Wien-filter, about which spins rotate as well

Concept of EDM measurement

- Determination of the invariant spin axis
- Deduce upper limit for deuteron EDM

E/B deflector development using real-scale lab setup

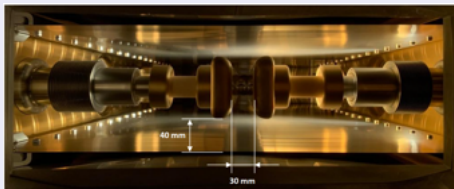


Equipment:

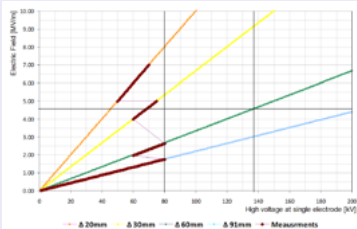
- Dipole magnet $B_{max} = 1.6$ T
- Mass = 64 t
- Gap height = 200 mm
- Protection foil between chamber wall and detector

Parameters:

- Electrode length = 1020 mm
- Electrode height = 90 mm
- Electrode spacing = 20 to 80 mm
- Max. applied voltage = ± 200 kV
- Material: Aluminum coated by TiN



- Electrodes at the distance of 30 mm inside the vacuum chamber



- Electric field between the electrodes vs displacement.

- Measurement procedure shortened due to time constraints.
- Max. electric field strength: 7 MV/m with 60 mm spacing between electrodes
- → Next step: setup moved to BNL?